

Logistic Regression and Confounding

180 minutes · 6 self-check questions

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Modelling binary outcomes: odds, odds ratios and what they do and do not mean, plus adjusting for confounding and recognising mediators and interactions.

Learning objectives

- Explain why linear regression is unsuitable for a binary outcome
- Interpret odds ratios and their confidence intervals correctly
- Distinguish odds ratio from risk ratio and know when they diverge
- Fit and interpret multivariable logistic regression
- Distinguish confounders, mediators and colliders, and adjust accordingly
- Assess model fit and predictive performance

Why not linear regression

- A binary outcome is 0 or 1, but a linear model can predict below 0 and above 1
- The residuals cannot be normal or homoscedastic when the outcome takes two values
- Logistic regression models the log odds, which is unbounded and well behaved
- The fitted probability is then obtained by transforming back through the logistic function
- Odds are $p / (1 - p)$: a probability of 0.2 corresponds to odds of 0.25
- The model is linear in the log odds, not in the probability — which is why effects are multiplicative

Reading an odds ratio

- $\exp(b)$ is the odds ratio for a one-unit increase in the predictor
- $OR = 1$ means no association; above 1 higher odds; below 1 lower odds
- The confidence interval is what matters: an OR of 3.2 with an interval of 0.8 to 12.6 is not evidence of much
- Odds ratios are not risk ratios. When the outcome is rare (under about 10%) they are close; when it is common they diverge sharply
- An OR of 3 for an outcome affecting 40% of people does not mean three times the risk

- Report the interval always, and consider a risk ratio or risk difference where the outcome is common and the design allows it
- In R: `fit <- glm(y ~ x, family = binomial, data = dat); exp(cbind(OR = coef(fit), confint(fit)))`

IN PRACTICE

The odds ratio is the coefficient logistic regression naturally produces, and it is often reported as though it were a risk ratio. For a common outcome that overstates the effect substantially — a mistake that appears in print constantly.

Confounders, mediators and colliders

- A confounder causes both the exposure and the outcome; adjusting for it removes bias
- A mediator lies on the causal path from exposure to outcome; adjusting for it removes part of the effect you are trying to measure
- A collider is caused by both exposure and outcome; adjusting for it creates bias where none existed
- The distinction cannot be made from the data — it comes from subject knowledge and the conceptual framework
- Draw a directed acyclic graph before choosing what to adjust for
- Adjusting for everything available is not conservative; it can actively introduce bias
- Example: adjusting smoking for lung function when estimating smoking's effect on mortality removes part of the very pathway of interest

WHAT TO DO WITH A THIRD VARIABLE

Role	Relationship	Adjust?	Effect of adjusting
Confounder	Causes exposure and outcome	Yes	Removes bias
Mediator	On the path exposure → outcome	Usually no	Removes part of the effect
Collider	Caused by exposure and outcome	No	Creates bias
Competing cause	Causes outcome only	Optional	Improves precision

Building and checking the model

- Rule of thumb: at least 10 events per predictor, counting the rarer outcome category, not the sample size
- 40 events and 8 predictors is over-fitted regardless of how large the sample looks
- Check linearity of continuous predictors on the log-odds scale, not the probability scale
- Hosmer-Lemeshow tests calibration, though it is sensitive to sample size and grouping
- The C-statistic (area under the ROC curve) measures discrimination: 0.5 is chance, above 0.8 is good

- Report both discrimination and calibration for a prediction model; either alone is incomplete
- Check for separation — a predictor that perfectly predicts the outcome produces enormous coefficients and unusable intervals

Reporting

- Report the number of events and the number of observations, not only n
- Present crude and adjusted estimates side by side
- State which variables were adjusted for and why they were chosen
- Give odds ratios with confidence intervals to two decimal places
- Say whether continuous predictors were entered linearly, categorised, or modelled with splines
- If the outcome is common, note that the odds ratio overstates the risk ratio

Worked examples

An odds ratio read as a risk ratio

A study of a common outcome — 40% of the unexposed group — reports $OR = 3.0$. The abstract says the exposure "triples the risk".

1. Odds in the unexposed: $0.40 / 0.60 = 0.67$.
2. Odds in the exposed: $0.67 \times 3.0 = 2.0$, so probability = $2.0 / 3.0 = 0.67$.
3. So risk goes from 40% to 67% — a risk ratio of about 1.7, not 3.0.
4. The odds ratio and risk ratio converge only when the outcome is rare, under roughly 10%.
5. At a 40% baseline they diverge substantially, and the abstract overstates the effect by nearly double.
6. Where the design allows, report a risk ratio or risk difference for a common outcome.

THE POINT

Logistic regression naturally produces odds ratios, and they are routinely described as risks. For common outcomes that is a real and large overstatement, and it appears in print constantly.

Adjusting for a mediator by accident

A study of smoking and mortality adjusts for age, sex, income and lung function. The smoking effect shrinks sharply, and the authors conclude smoking matters less than thought.

1. Age, sex and income plausibly cause both smoking and mortality — they are confounders, and adjusting is right.
2. Lung function is different: smoking damages it, and that damage causes mortality.
3. So lung function lies on the causal path from exposure to outcome — it is a mediator.
4. Adjusting for it removes the part of smoking's effect that operates through lung damage, which is most of it.
5. The adjusted estimate answers "the effect of smoking not acting through lung function", which is not the question asked.
6. Drawing a causal diagram before choosing covariates would have separated the confounders from the mediator.

THE POINT

Adjusting for everything available is not the cautious choice. Which variables to adjust for is a causal question answered from subject knowledge, not a statistical one answered from the data.

Key terms

Odds

The probability of an event divided by the probability of it not occurring.

Odds ratio

The ratio of odds between two groups; the natural output of logistic regression.

Risk ratio

The ratio of probabilities between two groups; diverges from the odds ratio when the outcome is common.

Confounder

A variable causing both exposure and outcome, biasing the crude association.

Mediator

A variable on the causal path between exposure and outcome.

Collider

A variable caused by both exposure and outcome; adjusting for it introduces bias.

C-statistic

Area under the ROC curve, measuring how well a model discriminates between outcomes.

Events per variable

The number of outcome events divided by the number of predictors, a guide to over-fitting.

Activity

For your own binary outcome, list every variable you might adjust for. Classify each as confounder, mediator, collider or competing cause, and justify the classification from what causes what — not from what the data show. Then state which you will adjust for.

Further reading

- **Hernán & Robins — Causal Inference: What If** — Free online. The clearest treatment of confounders, mediators and colliders.
- **Knol et al. (2012) — "What do case-control studies estimate?"** — On odds ratios versus risk ratios.
- **Peduzzi et al. (1996) — events per variable in logistic regression** — The origin of the ten-events-per-predictor guide.

Self-check questions

1. A study reports $OR = 3.0$ for an outcome occurring in 40% of the unexposed group. Reading this as "three times the risk" is:
 - a. Correct
 - b. An overstatement, because odds ratios exceed risk ratios when the outcome is common
 - c. An understatement
 - d. Correct only in case-control studies
2. You are estimating the effect of smoking on mortality and adjust for lung function. Lung function is most likely a:
 - a. Confounder, so adjustment is correct
 - b. Mediator, so adjusting removes part of the effect you are trying to measure
 - c. Collider
 - d. Competing cause
3. Your dataset has 500 participants, 40 of whom had the outcome, and you fit 8 predictors. The concern is:
 - a. The sample is too small overall
 - b. Only 5 events per predictor, so the model is over-fitted
 - c. Logistic regression cannot handle 8 predictors
 - d. The outcome is too common

4. An odds ratio of 2.4 with a 95% CI of 0.7 to 8.1 should be described as:
- A strong positive association
 - Consistent with anything from a modest protective effect to a large harmful one — the study is uninformative here
 - Evidence of no association
 - Statistically significant
5. A C-statistic of 0.52 indicates the model:
- Discriminates well
 - Discriminates barely better than chance
 - Is perfectly calibrated
 - Has too many predictors
6. Which is the correct R call for logistic regression?
- `lm(y ~ x, data = dat)`
 - `glm(y ~ x, family = binomial, data = dat)`
 - `aov(y ~ x, data = dat)`
 - `cor.test(y, x)`
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Answers

- b.** Odds and risk diverge as the outcome becomes common. With a 40% baseline, an OR of 3.0 corresponds to a risk ratio well under 2. The approximation only holds for rare outcomes.
 - b.** Smoking damages lung function, which contributes to mortality — it lies on the causal path. Adjusting for it answers a different question: the effect of smoking not acting through lung function.
 - b.** What limits a logistic model is the number of events, not the sample size. 40 events with 8 predictors is 5 per variable, below the usual guide of 10, giving unstable coefficients and intervals that are too narrow.
 - b.** The interval spans 1 and is very wide. The point estimate is not the finding; the interval is, and here it tells you the study could not pin the effect down.
 - b.** 0.5 is a coin flip and 1.0 is perfect discrimination. 0.52 means the model can hardly separate those who had the outcome from those who did not, whatever its p-values say.
 - b.** `glm` with `family = binomial` fits the logistic model. Omitting the family argument fits a linear model instead — which runs without error and gives quietly wrong results.
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